11 Dan invests £14 000 in a savings account.
The savings account pays compound interest at a rate of

4% for the first year
x% for each extra year.

There is a total amount of £15 297.10 in the savings account at the end of 3 years.

Work out the value of x.

After year 1: £14 000 × 1.04 = £14 560

1 mark
Apply the first-year rate.

$14\,560 \times (1 + x/100)^2 = 15\,297.10$

1 mark
The same unknown multiplier applies in the next TWO years.

$\text{Multiplier}^2 = 15\,297.10 \div 14\,560 = 1.050625$

$\text{Multiplier} = \sqrt{1.050625} = 1.025$

1 mark
Take the positive square root.

$x = (1.025 - 1) \times 100 = 2.5$

1 mark
The rate for each extra year is 2.5%.

$x =$ 2.5

(Total for Question 11 is 4 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE